



Course: Analog and Digital Communication
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 Total Marks: 100
 Weight: 50%
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Instruction/Notes: Attempt all parts of a question together.
 Fourier Transforms Table is provided.

Question # 1 [12+8=20 Marks]

[CLO: 2]

- a) For the signal $x(t)$, determine the essential bandwidth B Hz of $x(t)$ such that the energy contained in the spectral components of $x(t)$ of frequencies below B Hz is 95% of the signal energy E_x .

$$x(t) = \frac{4}{t^2 + 4}$$

- b) Find and draw the time autocorrelation function of the signal $g(t)$ shown in figure 1, and from it determine the ESD of $g(t)$.

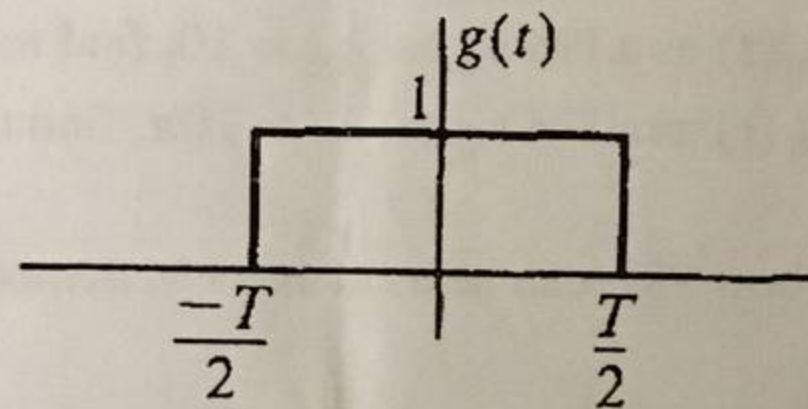


Figure 1

Question # 2 [10+10=20 Marks]

[CLO: 3]

- a) The signal $x(t) = \cos(200\pi t) + 3 \cos(400\pi t) + 2 \cos(800\pi t)$ is an input to a LSB-SC system with a carrier frequency of 10KHz.
- Sketch the LSB-SC spectrum of the transmitted signal. Find the bandwidth of the transmitted signal.
 - Can the block diagram shown in figure 2 be used to generate above mentioned LSB-SC signal? If no, please explain. If yes, find the range, and exact upper and lower frequencies of the band-pass filter.

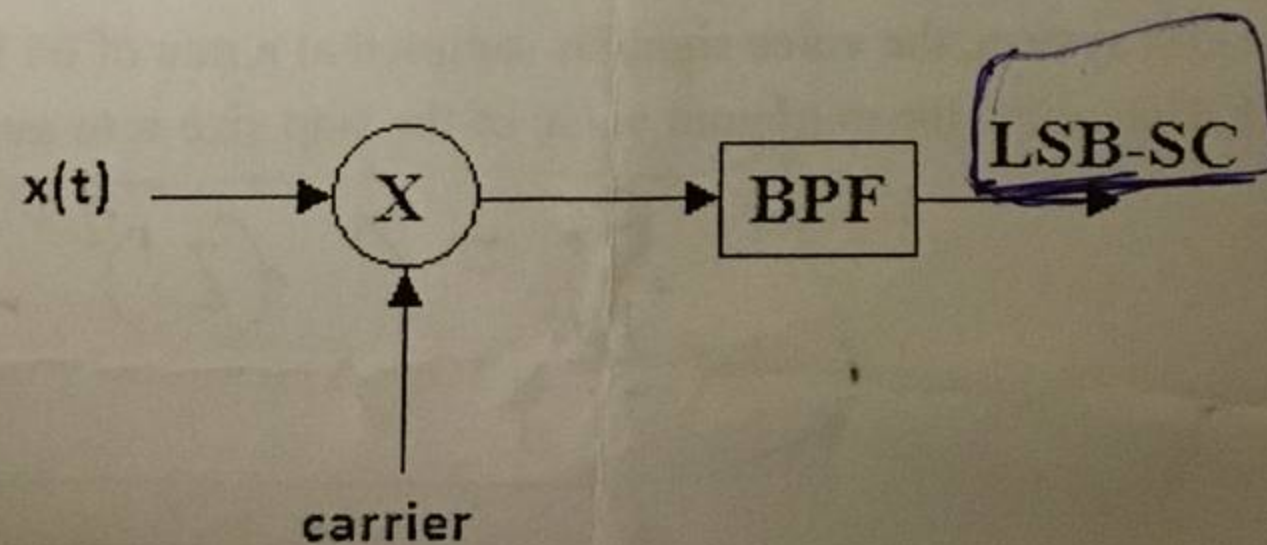


Figure 2

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

- b) System shown in figure 3 is used for scrambling audio signals. The output $z(t)$ is the scrambled version of the input $m(t)$. Where $\omega_0 = 20,000\pi$. Show whether or not we can descramble $z(t)$ to recover $m(t)$. Explain with proper reasoning and diagrams (spectrum) where required.

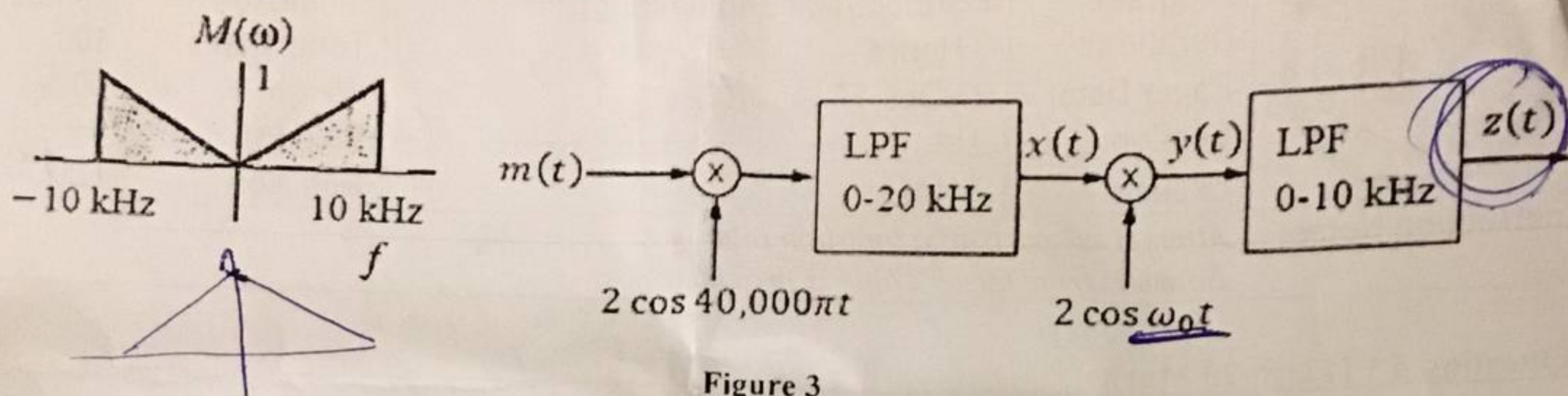


Figure 3

Question # 3 [15+5=20 Marks]

[CLO: 4]

- a) An angle modulated signal with carrier frequency $\omega_c = 2\pi \times 10^6$ is described by
- $$\phi_{EM}(t) = 10 \cos(\omega_c t + 0.1 \sin(10^3 \pi t))$$
- Find the power of the modulated signal
 - Find the frequency deviation Δf
 - Find the phase deviation $\Delta \phi$
 - Estimate the bandwidth of $\phi_{EM}(t)$
 - Considering $\phi_{EM}(t)$ as a PM signal $k_p = 10$, find $m(t)$
 - Considering $\phi_{EM}(t)$ as a FM signal $k_f = 10\pi$, find $m(t)$

- b) Explain mathematically how PLL can be used as a FM demodulator.

Question # 4 [15+5=20 Marks]

[CLO: 6]

- a) A satellite signal has a bandwidth of 15 KHz. This signal is sampled, quantized and binary-coded to obtain a PCM signal. Signal must be sampled at a rate 20% higher than the Nyquist rate. Message signal $m(t)$ is normalized to peak voltages of $\pm 1V$. The average message power equals to 20mW. The signal-to-quantization noise ratio (SQNR) is required to be 43dB.

- Determine the sampling rate.
- Determine the minimum number of bits required to encode a sample. 7-5648
- Determine the number of bits per second required to encode the signal.
- Determine the minimum bandwidth required to transmit this signal.

- b) In a single-integration DM system, the voice signal is sampled at a rate of 64 KHz. The maximum signal amplitude is $A_{max} = 1$. Determine the minimum value of the step size σ to avoid slope overload

$$\frac{3L^2 m_p^2}{(m_p)^2}$$

$$\frac{S_o}{N_o} = C \cdot (2)^{2n}$$

$$\omega_s = 2\pi \times 8000$$

↳ for voice signal

$$C \cdot (2)^{2n}$$

$$n = \frac{\log L}{\log 2}$$

$$2^n = L = 2^1$$

- a) Draw the **essential** block diagram of a typical digital communication system (DCS) starting from information source up to the sink with proper labeling.
- b) Explain the basis for deriving the likelihood ratio test

$$z(T) \underset{H_2}{\overset{H_1}{\leq}} \gamma$$

and under what conditions the above test would be the optimum decision making criteria.

- c) Binary data at a rate of 1Mbits/s is to be transmitted using Nyquist criterion pulse with $P(\omega)$ shown in **figure 4**. The frequencies f_1 and f_2 (in hertz) of this spectrum are adjustable. The cannel available for the transmission of this data has a bandwidth of 800KHz. Determine f_1 and f_2 and the roll-off factor.

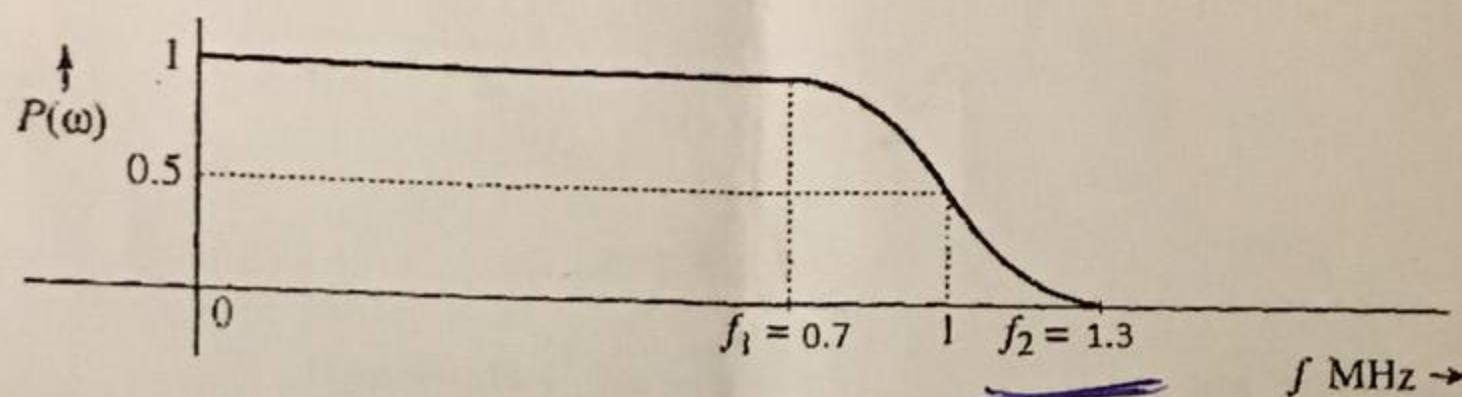


Figure 4

$P(f)$

FCSO CHM PBFM ← multiplex
Pulse → Detector

Short Table of Fourier Transforms

	$g(t)$	$G(\omega)$	
1	$e^{-at} u(t)$	$\frac{1}{a + j\omega}$	$a > 0$
2	$e^{at} u(-t)$	$\frac{1}{a - j\omega}$	$a > 0$
3	$e^{-a t }$	$\frac{2a}{a^2 + \omega^2}$	$a > 0$
4	$t e^{-at} u(t)$	$\frac{1}{(a + j\omega)^2}$	$a > 0$
5	$t^n e^{-at} u(t)$	$\frac{n!}{(a + j\omega)^{n+1}}$	$a > 0$
6	$\delta(t)$	1	
7	1	$2\pi \delta(\omega)$	
8	$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	
9	$\cos \omega_0 t$	$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	
10	$\sin \omega_0 t$	$j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$	
11	$u(t)$	$\pi \delta(\omega) + \frac{1}{j\omega}$	
12	$\text{sgn } t$	$\frac{2}{j\omega}$	
13	$\cos \omega_0 t u(t)$	$\frac{\pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + \frac{j\omega}{\omega_0^2 - \omega^2}$	
14	$\sin \omega_0 t u(t)$	$\frac{\pi}{2j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + \frac{\omega_0}{\omega_0^2 - \omega^2}$	
15	$e^{-at} \sin \omega_0 t u(t)$	$\frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$	$a > 0$
16	$e^{-at} \cos \omega_0 t u(t)$	$\frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$	$a > 0$
17	$\text{rect}\left(\frac{t}{\tau}\right)$	$\tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$	
18	$\frac{W}{\pi} \text{sinc}(Wt)$	$\text{rect}\left(\frac{\omega}{2W}\right)$	
19	$\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2} \text{sinc}^2\left(\frac{\omega\tau}{4}\right)$	
20	$\frac{W}{2\pi} \text{sinc}^2\left(\frac{Wt}{2}\right)$	$\Delta\left(\frac{\omega}{2W}\right)$	
21	$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$	$\omega_0 = \frac{2\pi}{T}$
22	$e^{-t^2/2\sigma^2}$	$\sigma\sqrt{2\pi} e^{-\sigma^2\omega^2/2}$	

$$\Delta(t)$$

Q#03

~~$$\Phi_{EM} = 2\pi(B + \Delta f t)$$~~

$$\begin{aligned} \sqrt{\Phi_{EM}} &= 2(\Delta f + B) \\ &= 2\left(50 + \frac{10^3 \pi}{2\pi}\right) \end{aligned}$$

solve karloa aagay

Q3(iv)

Q3(v)

$$\begin{aligned} \Phi_{PM} &= A \cos(\omega_c t + k_p m(t)) \\ &= A \cos(2\pi \times 10^6 t + 10 m(t)) \quad \text{--- (i)} \end{aligned}$$

mapping given signal on the above eq.

$$\Phi_{EM} = 10 \cos(\omega_c t + 0.1 \sin(10^3 \pi t)) \quad \text{--- (ii)}$$

equating eq. (i) & (ii) ; $A = 10$

$$10 m(t) = 0.1 \sin(10^3 \pi t)$$

$$m(t) = \frac{0.1}{10} \sin(10^3 \pi t)$$

Q#03

$$vi) \Phi_{FM} = A \cos \left[\omega_c t + k_f \int_{-\infty}^t m(\alpha) d\alpha \right]$$

$$\Phi_{FM}(t) = 10 \cos(\omega_c t + 0.1 \sin(10^3 \pi t))$$

equating both equations

$$k_f \int_{-\infty}^t m(\alpha) d\alpha = 0.1 \sin(10^3 \pi t)$$

$$10\pi \int_{-\infty}^t m(\alpha) d\alpha = 0.1 \sin(10^3 \pi t)$$

$$\int_{-\infty}^t m(\alpha) d\alpha = \frac{0.1}{10\pi} \sin(10^3 \pi t)$$

Differentiating both sides w.r.t "t"

$$m(t) = \frac{0.1 \times}{10\pi} \cos(10^3 \pi t) \times 10^3 \pi$$

$$m(t) = 10 \cos(10^3 \pi t)$$